CLINICAL EVIDENCE

Clinical evidence for olanzapine

Choose an indication below to see key results. Open a study for its design, safety findings and limitations.

Schizophrenia | Manic episodes | Recurrence prevention | Guidelines

Schizophrenia

Across five controlled trials, olanzapine improved both positive and negative symptoms. A published dose-ranging trial and a secondary analysis of mood symptoms add detail.

2 of 2

placebo-controlled trials: significantly greater improvement in negative and positive symptoms[1]

2 of 3

comparator-controlled trials: significantly greater improvement in negative and positive symptoms[1]

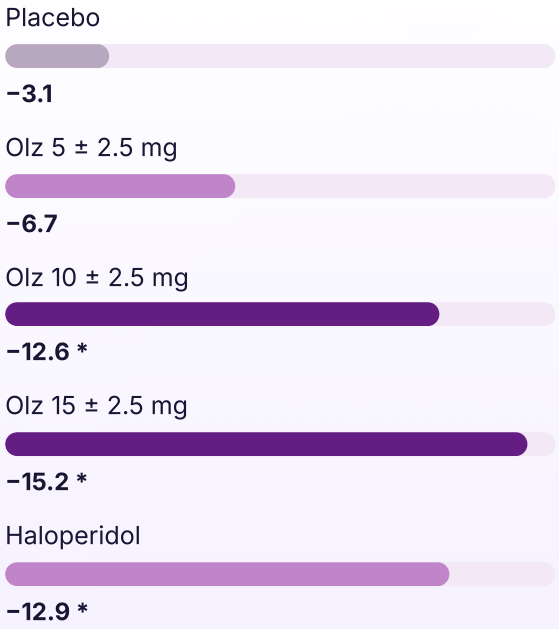
> 2,900

patients across these trials, with both positive and negative symptoms[1]

North American double-blind trial · Beasley et al. 1996 · 6 weeks · n = 335

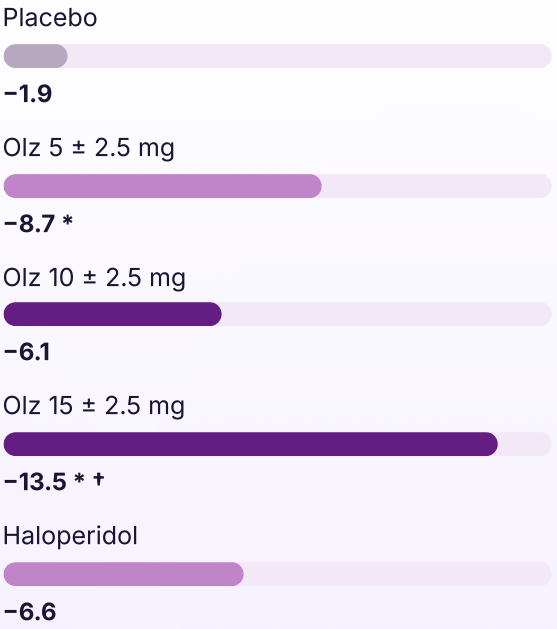
BPRS total −15.2 with olanzapine 15 ± 2.5 mg vs −3.1 with placebo.

RESULTS — BPRS TOTAL (OVERALL SYMPTOMS)



Mean change from baseline (LOCF). * significantly better than placebo.

RESULTS — SANS COMPOSITE (NEGATIVE SYMPTOMS)



* versus placebo. † also versus haloperidol.

In overall symptoms (BPRS total), the medium and high olanzapine dose ranges and haloperidol were superior to placebo.[10]

In negative symptoms (SANS composite), the low and high olanzapine ranges were superior to placebo, and the high range was also superior to haloperidol.[10]

* $p \leq 0.01$ versus placebo ($p \leq 0.05$ for SANS, olanzapine 5 ± 2.5 mg). † $p \leq 0.05$ versus haloperidol. Redrawn from Beasley et al. 1996, Table 5.[10]

STUDY DESIGN

DESIGN	Randomised, double-blind, placebo- and active-controlled; 22 sites, US and Canada
PATIENTS	n = 335; DSM-III-R schizophrenia, acute exacerbation; BPRS total ≥ 24; about 91% chronic course

ARMS	Olanzapine 5 ± 2.5, 10 ± 2.5 or 15 ± 2.5 mg/day; haloperidol 15 ± 5 mg/day; placebo
DURATION	6 weeks (acute phase)
ENDPOINTS	Change in BPRS total and BPRS positive versus placebo; SANS composite versus placebo and haloperidol (LOCF)[10]

SAFETY FINDINGS

39.1%
vs 16.2%

somnolence, olanzapine 15 ± 2.5 mg vs placebo

+3.5 kg

mean weight change at endpoint, olanzapine 15 ± 2.5 mg

0%
vs 13.0%

acute dystonia, olanzapine vs haloperidol

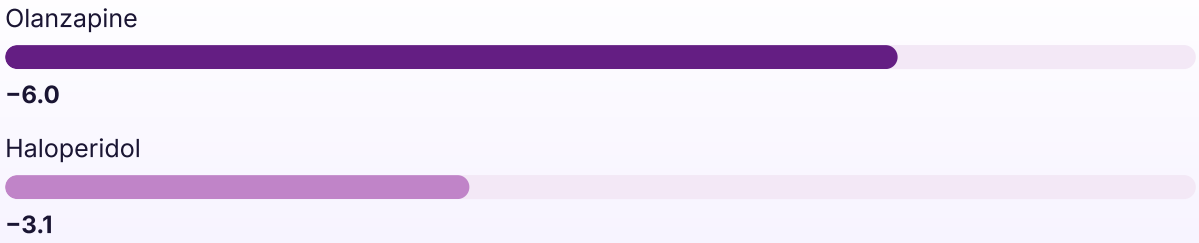
Dose-related transaminase elevations occurred; six olanzapine-treated patients discontinued because of raised ALT without clinical signs of hepatic dysfunction.[10]

LIMITATIONS

- Six-week acute phase only.
- High placebo response.
- The haloperidol dose (mean modal 16.4 mg/day) has been discussed as higher than optimal, which may favour comparisons against it.[10,8]

Mood symptoms in schizophrenia and related disorders · n = 1,481 · baseline MADRS 16.6

MADRS change −6.0 with olanzapine vs −3.1 with haloperidol (p = 0.001).



Mean change from baseline to endpoint. Bars represent improvement as scores fall. p = 0.001.

Patients had schizophrenia, schizoaffective disorder or related disorders, with varying degrees of depressive symptoms. A prospective secondary analysis of mood-score change favoured olanzapine.[1]

Limitation. This was a secondary analysis, not the primary outcome of the study.

Manic episodes

In moderate to severe manic episodes, olanzapine reduced manic symptoms versus placebo and divalproex, and when added to lithium or valproate.

48.6%
vs 24.2%

patients responding at 3 weeks, olanzapine vs placebo (p = 0.004)[11]

−10.3
vs −4.9

mean change in YMRS total at 3 weeks (p = 0.02)[11]

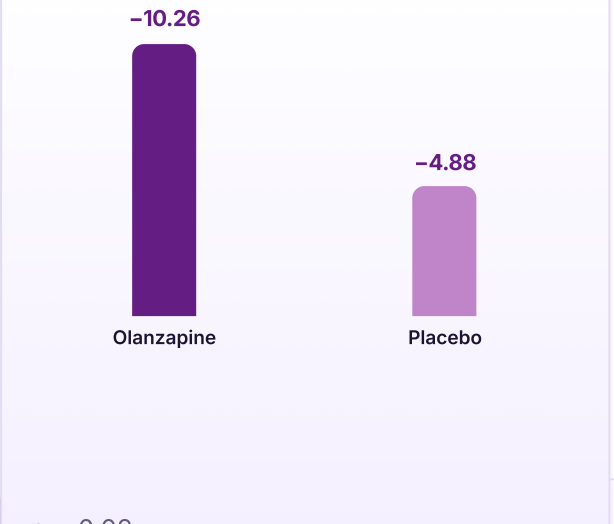
6 weeks

add-on to lithium or valproate: greater reduction in manic symptoms than either alone[1]

Olanzapine versus placebo in acute mania · Tohen et al. 1999 · 3 weeks · n = 139

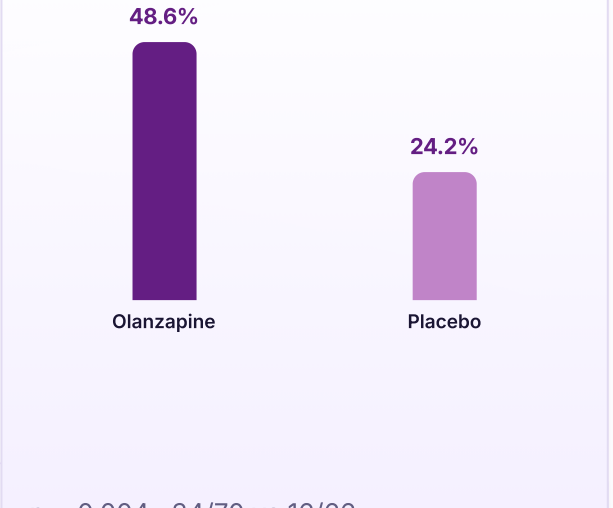
48.6% vs 24.2% of patients responded (p = 0.004).

RESULTS — MEAN CHANGE IN YMRS TOTAL



p = 0.02

RESULTS — RESPONDERS (≥ 50% YMRS DECREASE)



p = 0.004 · 34/70 vs 16/66

patients responded.[11]

Change from baseline to endpoint (LOCF) and proportion of patients with ≥ 50% decrease in YMRS at 3 weeks. Redrawn from Tohen et al. 1999.[11]

STUDY DESIGN

DESIGN	Randomised, double-blind, placebo-controlled, parallel group; 16 US sites
PATIENTS	n = 139 (olanzapine 70, placebo 69); DSM-IV bipolar I disorder, manic or mixed episode; YMRS ≥ 20; 53% with psychotic features
DOSING	Started at 10 mg/day, adjustable 5–20 mg/day; mean modal dose 14.9 mg/day
DURATION	3 weeks
ENDPOINTS	Primary: change in YMRS total (LOCF). Response defined in advance as ≥ 50% YMRS decrease [11]

SAFETY FINDINGS

32.9%
vs 17.4%

somnolence, olanzapine vs placebo

+1.65 kg
vs −0.44

mean weight change over 3 weeks

17.6%
vs 0%

treatment-emergent ALT elevations

Dry mouth (25.7% vs 8.7%), dizziness (22.9% vs 5.8%) and weight gain reported as an adverse event (11.4% vs 1.4%) were also more frequent with olanzapine.[11]

LIMITATIONS

- Three-week duration.
- High placebo dropout (65.2%).
- Concomitant lorazepam permitted.
- The trial started at 10 mg/day; the recommended starting dose for monotherapy of a manic episode is 15 mg/day.[11,1]

Further controlled studies in manic or mixed episodes · 3 to 12 weeks

Versus placebo · 3 weeks. Superior efficacy in reducing manic symptoms.[1]

Versus divalproex · 3 weeks. Superior efficacy to valproate semisodium (divalproex) in reducing manic symptoms.[1]

Versus haloperidol · 6 and 12 weeks. Comparable proportion of patients in symptomatic remission from mania and depression.[1]

Add-on to lithium or valproate · 6 weeks. In patients already treated for at least 2 weeks, adding olanzapine 10 mg reduced manic symptoms more than lithium or valproate alone.[1]

Effect sizes are not available for these studies, so results are described in words.

Recurrence prevention

In patients whose manic episode responded, continued olanzapine was superior to placebo in preventing recurrence and non-inferior to lithium. Added to lithium or valproate, it was not superior to either alone.

Superior

to placebo on bipolar recurrence over 12 months[1]

30.0%
vs 38.3%

recurrence vs lithium over 12 months: non-inferior (p = 0.055)[1]

Not superior

as 18-month co-therapy, versus lithium or valproate alone[1]

Continued olanzapine versus placebo · Remission on olanzapine · 12 months

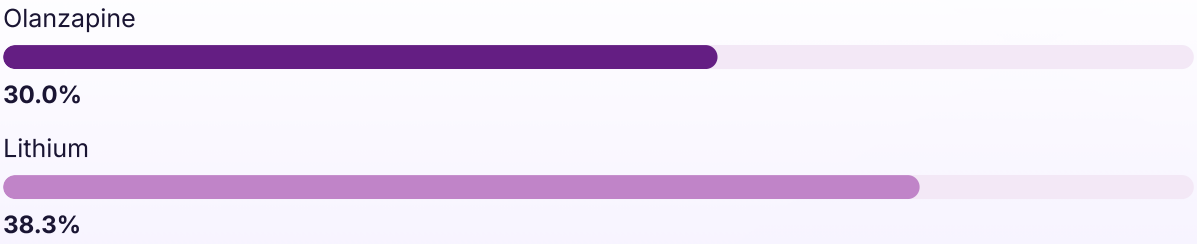
Statistically significantly superior to placebo on bipolar recurrence.

Design. Patients with a manic episode who achieved remission on olanzapine were randomised to olanzapine or placebo for 12 months.

Result. Olanzapine was statistically significantly superior to placebo on the primary endpoint of bipolar recurrence, and also showed an advantage in preventing recurrence into either mania or depression.[1]

Recurrence rates are not available for this study, so no chart is shown.

Olanzapine versus lithium · 12 months · recurrence 30.0% vs 38.3%



Patients with bipolar recurrence at 12 months (primary endpoint). Olanzapine was statistically non-inferior to lithium, p = 0.055.

After remission on olanzapine plus lithium, patients were randomised to olanzapine alone or lithium alone. Olanzapine was statistically non-inferior to lithium on the primary endpoint of bipolar recurrence.[1]

Olanzapine added to lithium or valproate · 18 months

In patients with a manic or mixed episode stabilised on olanzapine plus lithium or valproate, continuing olanzapine co-therapy was not statistically significantly superior to lithium or valproate alone in delaying bipolar recurrence, defined by syndromic (diagnostic) criteria.[1]

Guidelines

Guidelines place olanzapine among several options rather than ahead of them. Recommendations depend on the phase of illness, patient preference and how well the adverse-effect profile suits the individual.

APA practice guideline, 2021[6]

SCHIZOPHRENIA

- Treat with an antipsychotic and monitor effectiveness and side effects (1A).
- Continue antipsychotic treatment after symptoms improve (1A), preferably with the same medicine (2B).
- Clozapine for treatment-resistant schizophrenia (1B).
- Long-acting injectable if preferred or adherence is poor (2B).
- Individual antipsychotics are not ranked.

BAP guidelines, 2016[7]

BIPOLAR DISORDER

- Severe mania: haloperidol, olanzapine, risperidone and quetiapine described as particularly effective.
- Long term: olanzapine is among the agents shown to prevent manic relapse; lithium is recommended as initial monotherapy.
- Weight gain is a prominent adverse effect of olanzapine.

WPA review, 2008[8]

SCHIZOPHRENIA

- Apart from clozapine, no consistent efficacy differences between second-generation antipsychotics.
- Olanzapine is associated with more weight gain and metabolic adverse effects than most comparators.

WHO Essential Medicines List, 2023[9]

PSYCHOTIC AND BIPOLAR DISORDERS

- Olanzapine listed as a therapeutic alternative to risperidone (psychotic disorders) and to quetiapine (bipolar disorders).

Reading this evidence

Consider each result in the context of its population, comparator, endpoint and duration. Individual study results do not predict the response of every patient.

All references →

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Next screen: **References** →

Address

CHEPLAPHARM Registration GmbH
Weiler Straße 5e
79540 Lörrach, Germany

Contact

+49 7621 593 06-00

+49 7621 593 06-98

info@cheplapharm-reg.com

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